

Applicability of variogram-based Gaussian process regression to industrial flotation data: a copper–molybdenum case study

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Abstract

Variogram-based Gaussian process regression has recently been demonstrated as a probabilistic framework for optimising mineral processing circuits, with validation on a statistically designed flotation campaign. Its applicability to plant campaign data, which are typically acquired one factor at a time and are subject to measurement noise, has not been examined. The present study applies the published workflow, without modification, to thirty-eight flotation tests from a copper–molybdenum concentrator. Predicted recoveries outside the physically admissible range are obtained over a substantial fraction of the interpolated domain, and the reported Pareto frontier is not reproducible under posterior resampling. Three modelling assumptions account for these observations, and modifications to each are proposed and evaluated by leave-one-out cross-validation. The modified workflow yields predictions that satisfy physical bounds, an uncertainty estimate consistent with the observed measurement variability, and an operating window that agrees with independent conclusions drawn from the same campaign.

Keywords: froth flotation, Gaussian process regression, uncertainty quantification, multi-objective optimisation, copper–molybdenum separation

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1. Introduction

The progressive depletion of high-grade, readily processed sulphide ores has increased the sensitivity of flotation performance to variations in feed mineralogy and pulp chemistry, narrowing the range of operating conditions over which target grades and recoveries can be maintained simultaneously. Copper–molybdenum separation illustrates the difficulty well, since molybdenite and chalcopyrite are both naturally hydrophobic and the reagent regimes that promote the flotation of one generally promote the flotation of the other as well.

Surrogate models that predict metallurgical response from operating variables, and that quantify the confidence associated with each prediction, are consequently of practical value. Gaussian process regression is well suited to this setting: it performs comparatively well on small datasets and provides a posterior variance as an intrinsic part of its output rather than as a separate estimate.

The specification of the prior covariance function is a central question in such applications. Eskanolu, Yin and Caers proposed deriving it directly from the data through empirical variograms fitted in the space of the operating variables, and demonstrated the approach on a designed flotation campaign, reporting a Pareto-efficient frontier between two competing responses which could be mapped back to process settings. The formulation is attractive because the covariance structure is estimated rather than assumed, and the accompanying implementation has been made openly available.

The present work examines how this framework behaves when it is applied to data of a different character. Designed experiments distribute observations through the predictor space and are conducted under controlled conditions; plant campaign data are acquired one factor at a time, populate the space along coordinate directions, and carry measurement variability that a designed programme is intended to minimise. Whether the framework transfers between these two regimes is a question of practical consequence for industrial application, and it is the question addressed here.

2. Data

The dataset originates from a molybdenum concentrator at Jinduicheng, Shaanxi Province, China, where chalcopyrite occurs as a penalty impurity in the molybdenite concentrate. Seasonal variation in the copper content of the concentrate motivated a two-year experimental campaign, in which grinding fineness, pulp pH, diesel collector dosage and diesel–pulp conditioning time

were varied while molybdenum and copper recovery were measured on the rougher concentrate. Thirty-eight tests were available for analysis; the ranges of the variables and responses are summarised in Table 1.

Table 1: Operating variables and metallurgical responses measured on the rougher concentrate.

	Variable	Range	Unit
x_1	Grinding fineness (-0.074 mm)	40–75	%
x_2	Pulp pH	3.5–11	–
x_3	Diesel dosage	50–200	g/t
x_4	Conditioning time	5–25	min
y_1	Mo recovery	7–58	%
y_2	Cu recovery	2–52	%

Two characteristics of the data acquisition are relevant to the analysis that follows. The campaign was conducted in a one-factor-at-a-time manner, which is the usual practice for plant trials, with the consequence that the thirty-eight observations are distributed along a limited number of coordinate directions rather than throughout the interior of the four-dimensional predictor space. In addition, pulp pH was varied in only one of the test series, so that for the greater part of the dataset this variable remains constant.

3. Behaviour of the reference workflow

The published procedure comprises the following steps: the predictors are scaled to the unit interval; the standardised responses are reduced to two principal components; an empirical variogram is fitted to each component as a function of the predictor coordinates; ordinary kriging is performed; an ensemble of conditional realisations is generated on an augmented grid; each realisation is transformed back through the inverse principal component and standardisation steps; and the Pareto frontier of the ensemble mean surface is computed.

Three elements of this sequence are adopted in the original work without accompanying sensitivity analysis, and each is examined below with respect to the present dataset.

The first concerns the treatment of the principal components. Independently simulated realisations of PC_1 and PC_2 are combined through the inverse principal component transformation. Since PCA ensures that the components are uncorrelated but not that they are independent, and since

conditional simulation is a nonlinear operation, this combination is not generally correlation-preserving. For the present dataset, the resulting responses lie considerably outside the convex hull of the observations; together with an interpolation grid that extends beyond the data bounds in every scaled coordinate, this produces the predictions shown in Fig. 1.

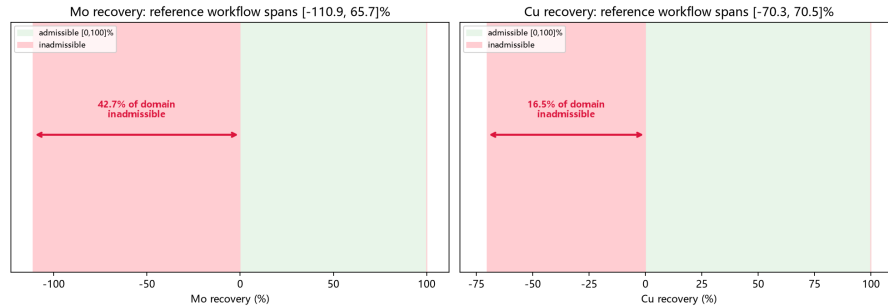


Figure 1: Application of the reference workflow to the industrial dataset. The shaded region indicates the range physically accessible to a recovery value; the predictions obtained for molybdenum and copper recovery fall outside this range over 42.7% and 16.5% of the interpolated domain, reaching -110.9% and -70.3% respectively.

The second concerns the covariance structure. A single length scale is applied to all four predictors, which implies that each influences the response equally per unit of scaled distance. This equality refers to the numerical scaling of the variables rather than to their relative metallurgical significance, and it is the contrast between the two that the subsequent analysis shows to be of interest.

The third concerns interpolation. Kriging is performed such that the surface reproduces the observations exactly, which is appropriate for a deterministic simulator and is not appropriate where measurement variability is present. When the noise level is estimated from the data rather than constrained to zero, values of 0.64 and 0.46 in standardised units are obtained for the two responses, corresponding to approximately half the response standard deviation. The reference workflow reports zero posterior variance at these same measurement locations. It may further be noted that the original procedure reports no cross-validation or held-out error, so that discrepancies of this kind are not detectable from its outputs alone.

4. Modifications and their effect

Four changes were made. The first three address the assumptions identified above, and the fourth concerns the decision problem itself.

The responses were modelled directly rather than through principal components, which removes the inverse transformation from the prediction pathway. One process was fitted per response; this does not retain the correlation between responses, and a co-regionalised formulation that would do so is identified below as the natural extension, but the predictions become physically admissible as a result.

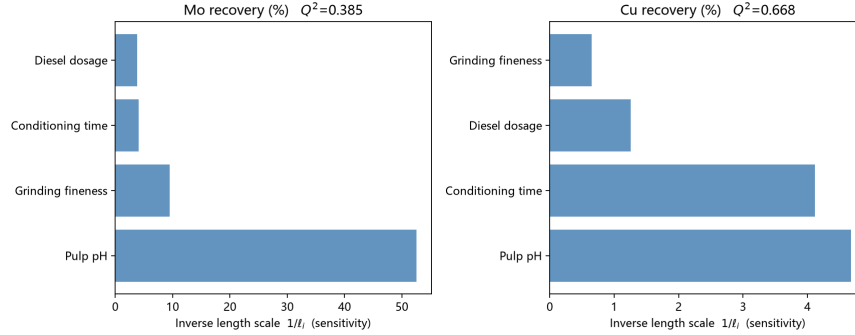


Figure 2: Inverse length scales, interpreted as predictor sensitivities, for each response. The two profiles differ both in magnitude and in ordering.

An automatic relevance determination covariance function was adopted,

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left(-\frac{1}{2} \sum_{i=1}^d \frac{(x_i - x'_i)^2}{\ell_i^2}\right) + \sigma_n^2 \delta_{xx'}, \quad (1)$$

with all hyperparameters determined by maximising the marginal likelihood. The inverse length scales are then interpretable as sensitivities, and are shown in Fig. 2. Molybdenum recovery is dominated by pulp pH, which is five and a half times more influential than the next variable and thirteen times more influential than grinding fineness, whereas copper recovery exhibits a flatter profile ordered differently, with pH and conditioning time of comparable importance. A covariance function with a single length scale cannot represent either of these structures.

Kriging was performed in non-exact form with the nugget estimated from the data, which allows the model to represent the finite reproducibility of flotation testing and permits the predictive performance to be quantified. Leave-one-out cross-validation yields coefficients of determination of 0.39 and 0.67 for molybdenum and copper recovery, with corresponding RMSE values of 10.6 and 5.9 percentage points (Fig. 3). The difference between the two is of interest: molybdenum recovery evidently depends on variables not

represented among the four predictors, while copper recovery is more nearly determined by them.

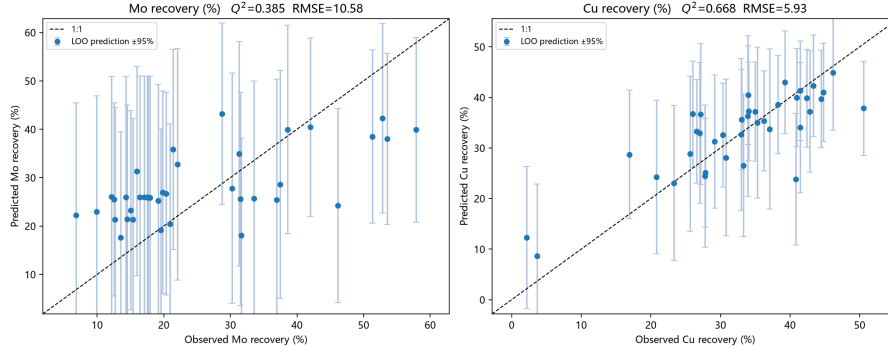


Figure 3: Leave-one-out predictions against observed values, with 95% credible intervals.

The fourth modification concerns the objective formulation. Principal component loadings of $[0.707, 0.707]$ indicate that the two recoveries are positively correlated over the tested conditions, so that the efficient frontier between them is short and describes limited trade-off. To assess the stability of the frontier, four hundred realisations were sampled from the posterior and the frontier recomputed for each. The mean-field frontier contains five points, and across the four hundred realisations no point remains on the frontier in more than ten percent of them. A chance-constrained dominance criterion was also examined, and returned an empty set at all confidence levels tested. These observations indicate that the Pareto formulation is not well-posed for the present response pair.

An alternative formulation was therefore adopted, in which the operating conditions are sought for which the probability of exceeding specified recovery targets simultaneously is acceptably high. Taking the targets at the sixtieth percentile of the observations, corresponding to 23% molybdenum and 35% copper recovery, the feasible set comprises forty-four grid points at fifty percent confidence and eight at seventy-five percent. At ninety percent confidence the set is empty, which indicates that the target combination is demanding given the variability of the circuit, rather than that the analysis has failed. The most favourable point corresponds to a grinding fineness of approximately 70%, pulp pH of 7.0, diesel dosage of 100 g/t and conditioning time of 15 min, with predicted recoveries of $45.7 \pm 9.9\%$ for molybdenum and $41.3 \pm 5.0\%$ for copper at a joint confidence of 0.88. The conditioning time and collector dosage obtained are consistent with the conclusions of

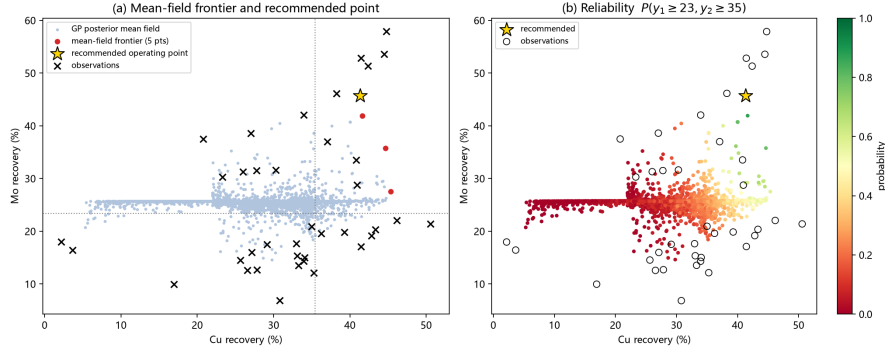


Figure 4: (a) Posterior mean field with the mean-field frontier and the recommended operating point. (b) Probability of simultaneously exceeding both recovery targets.

the original campaign, which the reference workflow was not in a position to verify.

5. Discussion

The observations reported here are not specific to flotation. Variogram-based methods were developed for spatial data, in which the distance between two samples is a physical quantity and the variogram summarises the spatial continuity of the phenomenon. When the same formalism is applied in a parameter space, distance becomes a modelling choice rather than a physical one. For a designed experiment, in which observations are distributed through the space, the consequences of that choice are limited. For one-factor-at-a-time plant data, the observations lie along coordinate directions, inter-point distances are dominated by whichever variable was varied, and an isotropic covariance function attributes this geometrical property to physical similarity.

Several diagnostics follow, none of which require prior knowledge of the correct result. The number of observations per effective dimension can be counted, the extent of the interpolation grid relative to the data can be inspected, it can be verified that predictions respect bounds known to apply to the quantity of interest, and the stability of derived quantities under posterior resampling can be assessed. For the present dataset, three of these four would have identified the issues described above before they appeared in any figure.

Two limitations should be noted. The dataset comprises thirty-eight observations from a single deposit, and the behaviour described is demonstrated on one case rather than established statistically; the accompanying code

permits the same diagnostics to be applied to other datasets. In addition, modelling the responses independently forgoes the correlation between them, which the data contain. A co-regionalised formulation would retain this information and is considered the most valuable immediate extension, particularly in view of the fact that the operational decision at this concentrator concerns concentrate grade rather than recovery, where the two objectives are in genuine conflict and a frontier formulation would be well posed.

6. Conclusions

A variogram-based Gaussian process workflow that performs satisfactorily on a designed flotation experiment does not transfer without modification to plant campaign data. Three assumptions underlie this difference, and each is identifiable from the data in advance: independent simulation of principal components which are subsequently combined through a nonlinear inverse transformation, a single correlation length applied across predictors of unequal influence, and exact interpolation of observations subject to measurement variability. Modifying each of these, and replacing the Pareto frontier with a reliability-based criterion once the responses were found not to be in conflict, produces predictions that satisfy physical bounds, an uncertainty estimate consistent with the observed variability, and an operating window in agreement with the independent conclusions of the campaign. These modifications are recommended as prerequisites for the application of variogram-based Gaussian process regression to industrial flotation data.